



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

REVIEW TEST

17-10-2021

JEE(Main + Advanced) : NURTURE COURSE [PHASE : TEPS, TOPS, I(A), I, I(B), II(A), II & LIVE-I, I(B), II(A)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	D	C	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,C	B,C	B,C	A,B	A,B,C,D	A,B,C	A,B,C	A,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	0.14	130.00	3.00	0.66 to 0.67	2.40	6.00		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	C	C	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,C,D	A,B	A,C,D	B,C,D	B,D	A,C	A,B,C	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	6.00	5.00	6.00	600.00	5.00	6.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	B	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,D	A,C	A,D	B,C,D	B,C	A,B,D	A,C,D	B,C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	1.00	7.00	6.00	4.00	7.00	4.23 or 4.24		

HINT – SHEET

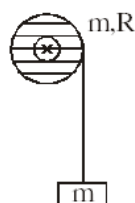
PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)

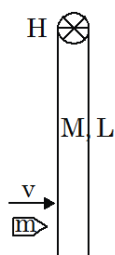
$$a = \frac{mg}{M + I/R^2}$$

$$a = \frac{2g}{3}$$

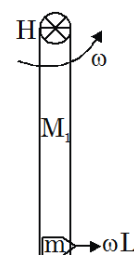


2. Ans (D)

just before collision



just after collision



$$(L_i)_H = (L_f)_H$$

$$mvL(\hat{v}) = m(\omega L)(L)\hat{v} + \frac{ML^2\omega}{3}\hat{v}$$

$$mv^2 = \left(\frac{3m + M}{3}\right)\omega v^2$$

$$\omega = \frac{3mv}{(3m + M)L}$$

3. **Ans (C)**

No horizontal thrust, velocity of trolley remain constant and horizontal momentum conserved.

4. **Ans (C)**

Energy conservation

$$\frac{1}{2}K\left(\frac{\ell_0}{2}\right)^2 + \frac{1}{2}mv'^2 = \frac{1}{2}mv_0^2 \quad \dots(i)$$

& Angular momentum conservation

$$m \cdot v_0 \ell_0 = mv' \left(\ell_0 + \frac{\ell_0}{2} \right) \quad \dots(ii)$$

From equation (i) & (ii) :

$$v_0 = \frac{3\ell_0}{2} \sqrt{\frac{k}{5m}}$$

SECTION-I (ii)

6. **Ans (B,C)**

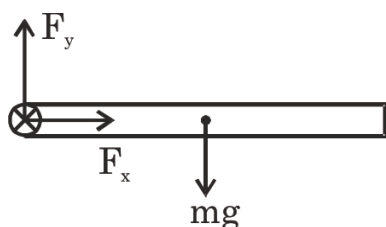
Whole mass can be assumed at COM for calculation of gravitational PE and linear momentum.

8. **Ans (A,B)**

When rod becomes horizontal
WET

$$mg \frac{\ell}{2} \sin \theta = \frac{1}{2} \left(\frac{m\ell^2}{3} \right) \omega^2 - 0$$

$$\omega^2 = \frac{3g \sin \theta}{\ell}$$



$$\text{and } T_0 = mg \frac{\ell}{2} = \frac{m\ell}{3} \alpha$$

$$\alpha = \frac{3g}{2\ell}$$

$$\vec{a}_{in} = -\omega^2 \frac{\ell}{2} \hat{i} + \left(-\frac{\ell}{2} \alpha \right) \hat{j}$$

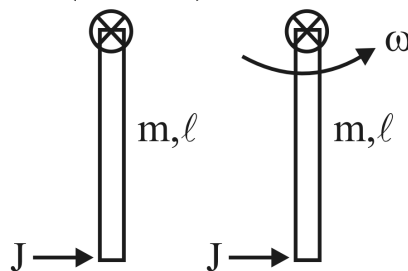
$$\vec{a}_{in} = \frac{-3g \sin \theta}{2} \hat{i} - \frac{3g}{4} \hat{j}$$

$$F_x = -m \left(\frac{3g \sin \theta}{2} \right)$$

$$F_y = mg - F_y = m \left(\frac{3g}{4} \right)$$

$$F_y = \frac{mg}{4}$$

9. **Ans (A,B,C,D)**



Initial Final

(A) Angular Impulse = ΔL

$$J \times \ell = L_f - L_i \quad [L_i = 0]$$

$$= L_f = J\ell$$

(B) $L_f = J\ell$

$$I_{axis} \omega = J\ell = \frac{m\ell^2 \omega}{3} = J\ell$$

$$\Rightarrow \omega = \frac{3J}{m\ell} \quad \dots(B)$$

(C) Kinetic energy

$$\begin{aligned} &= \frac{1}{2} I \omega^2 = \frac{1}{2} \frac{m\ell^2}{3} \times \left(\frac{3J}{m\ell} \right)^2 \\ &= \frac{1}{2} \frac{m\ell^2}{3} \times \frac{9J^2}{m^2 \ell^2} = \frac{3J^2}{2m} \quad \dots(C) \end{aligned}$$

(D) Linear Velocity of mid point of rod :

$$V = \omega r$$

$$r = \ell/2$$

$$v = \frac{\omega \ell}{2}$$

$$v = \frac{3J}{m\ell} \left(\frac{\ell}{2} \right)$$

$$v = \frac{3J}{2m} \quad \dots(D)$$

All Options are Correct

10. **Ans (A,B,C)**

conceptual

12. **Ans (A,C)**

COAM about hinge

$$M \frac{v}{2} \cdot \frac{\ell}{2} = \left(m \frac{\ell^2}{3} + m \frac{\ell^2}{4} \right) \omega$$

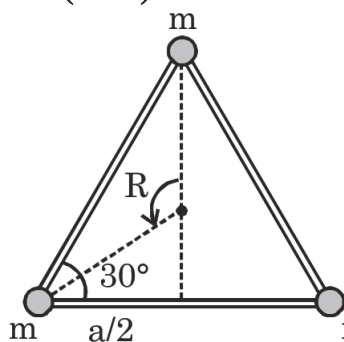
$$\frac{v}{4} = \frac{7}{12} \ell \omega \Rightarrow \omega = \frac{3v}{7\ell} \dots (i)$$

$$\text{COE } \frac{1}{2} \cdot \frac{7}{12} \omega \ell^2 = 2mg \frac{\ell}{2} \Rightarrow \omega = \sqrt{\frac{24g}{7\ell}} \dots (ii)$$

$$\text{from equation (i) and (ii) } v = \sqrt{\frac{168}{9}} g \ell$$

SECTION-II

1. Ans (0.14)



$$KE_{\text{rot}} = \frac{1}{2} I \omega^2 = \frac{1}{2} 3m \left(\frac{a}{\sqrt{3}} \right)^2 \omega^2 = \frac{1}{2} m a^2 \omega^2$$

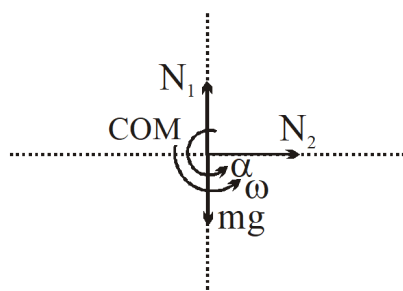
$$= \frac{1}{2} \times \frac{1}{5} \times \frac{4}{25} \times 9 = \frac{18}{125} = 0.14$$

2. Ans (130)

$$N_1 - mg = \frac{m\omega^3 \ell}{4} \quad \dots(i)$$

$$N_2 = m \left(\frac{\alpha \ell}{4} \right) \quad \dots(ii)$$

$$\tau = I\alpha \Rightarrow a = 0 \Rightarrow N_2 = 0$$



Using E/C

$$\frac{mg\ell}{4} = \frac{1}{2} \left(\frac{m\ell^2}{12} + \frac{m\ell^2}{16} \right) \omega^2$$

$$\Rightarrow \omega^2 = \frac{24g}{7\ell}$$

from eq. (i)

$$\Rightarrow N_1 = mg + m \left(\frac{6g}{7\ell} \right)$$

$$m = 7 \text{ kg}, \ell = 1$$

$$\Rightarrow N_1 = 130 \text{ N}$$

3. Ans (3)

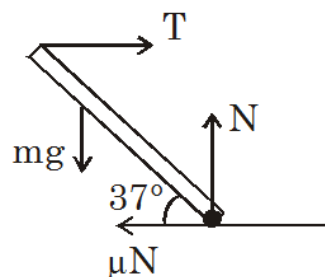
$$\tau_{\text{cm}} = 0$$

$$mg \sin \theta \times \frac{a}{2} = mg \cos \theta \times x$$

$$x = \frac{a}{2} \tan \theta = \frac{8}{2} \times \frac{3}{4} = 3 \text{ m}$$

4. Ans (0.66 to 0.67)

T the FBD of any one rod is



$$T = \mu N \quad \dots(i)$$

$$mg = N \quad \dots(ii)$$

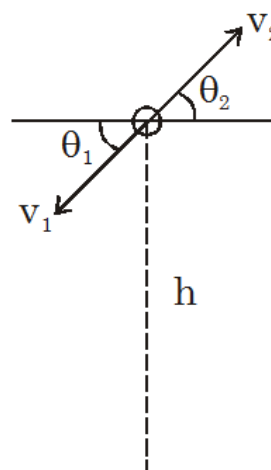
Taking torque of any one rod is

$$mg \frac{L}{g} \cos 37^\circ = TL \sin 37^\circ$$

$$T = \frac{mg}{2} \frac{4}{3} = \frac{2mg}{3}$$

$$\frac{2mg}{3} = \mu mg \Rightarrow \mu = \frac{2}{3}$$

5. Ans (2.40)



Bomb explode at highest point

$$\left. \begin{matrix} p_i = 0 \\ p_f = 0 \end{matrix} \right\} \text{ just before and after explosion.}$$

Since both travel same horizontal distance $v_1 \cos \theta_1 t_1$

$$= v_2 \cos \theta_2 t_2$$

$$\frac{v_1 \cos \theta_1}{v_2 \cos \theta_2} = \frac{t_2}{t_1}$$

By momentum conservation $m_1 v_1 \cos \theta_1 = m_2 v_2 \cos \theta_2$

$$\frac{v_1 \cos \theta_1}{v_2 \cos \theta_2} = \frac{m_2}{m_1} = 2 = \frac{t_2}{t_1}$$

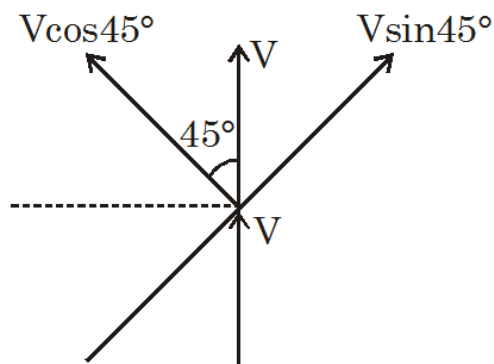
$$t_2 = 2t_1 \text{ \& } t_2 - t_1 = 4$$

$$t_1 = 4 \quad t_2 = 8$$

$$\frac{t_1 + t_2}{5} = \frac{12}{5} = 2.40$$

6. Ans (6.00)

Just before collision



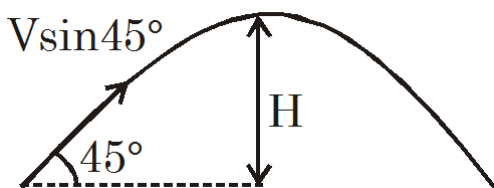
$$V^2 = (10\sqrt{3})^2 + 2(-10)(3)$$

$$= 300 - 60$$

$$V^2 = 240$$

$$V = \sqrt{240}$$

After collision



$$\Rightarrow H_{\max} = \frac{u^2 \sin^2 \theta}{2g} = \frac{(v \sin 45^\circ)^2 \sin^2 45^\circ}{2g}$$

$$= \frac{V^2 \sin^4 45^\circ}{2g}$$

$$= \frac{240}{20} \times \frac{1}{4}$$

$$= 3 \text{ m}$$

$$\Rightarrow H_{\text{total}} = 3 + 3 \text{ m}$$

$$H = 6 \text{ m}$$

PART-2 : CHEMISTRY

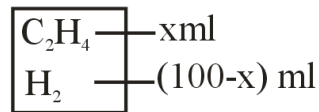
SECTION-I (i)

3. Ans (C)

$$\text{moles} = \frac{6.2}{62} \times (32e^-)$$

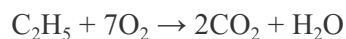
SECTION-I (ii)

8. Ans (B,C,D)



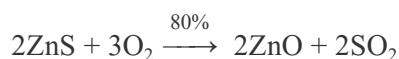
$$VC = 40 - 100 - [x + 100 - x - x]$$

$$x = 40$$

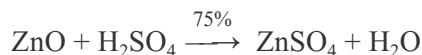


$$80 \text{ ml}$$

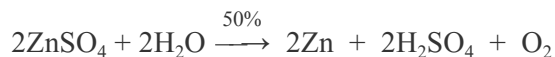
9. Ans (B,D)



$$15 \qquad 15 \times 0.8 = 12$$



$$12 \qquad 12 \times 0.75 = 9$$



$$\begin{array}{cccc} 9 & 9 \times 0.5 & 9 \times 0.5 & 4.5 \times 0.5 \\ & = 4.5 & = 4.5 & = 2.25 \end{array}$$

12. Ans (A,B,C,D)

$$N > P : \text{IE order}$$

$$\begin{array}{cc} \downarrow & \downarrow \\ 2^{\text{nd}} & 3^{\text{rd}} \text{ period} \end{array}$$

$$\left(\text{IE} \propto \frac{1}{\text{size}} \right)$$

$$O^+ > N^+ : \text{IE order}$$

$$(2p^3) \quad (2p^2)$$

$$Cl > B : \text{EN order}$$

SECTION-II

3. **Ans (6.00)**



$$x = 4, z = 6, p = 4$$

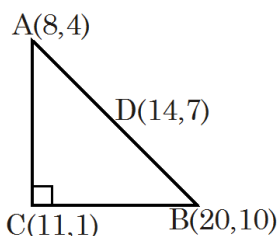
6. **Ans (6.00)**

$$\text{PM} = dRT, d \propto \frac{1}{T}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B)**



ΔABC right angle triangle

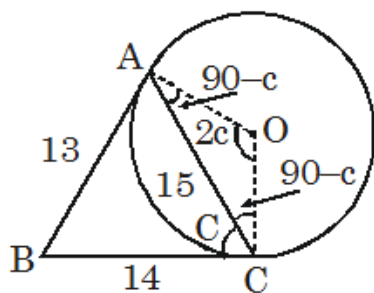
$$\angle C = 90^\circ$$

orthocentre = C

circumcentre = D

$$CD = \sqrt{9+36} = \sqrt{45}$$

2. **Ans (A)**



Using sine rule in ΔAOC ,

$$\frac{15}{\sin 2C} = \frac{r}{\cos C} \Rightarrow r = \frac{15}{2 \sin C}$$

$$\text{Now } \cos C = \frac{15^2 + 14^2 - 13^2}{2 \cdot 15 \cdot 14} = \frac{9}{15}$$

$$\Rightarrow \sin C = \frac{12}{15} \quad r = \frac{15 \cdot 15}{2 \cdot 12} = \frac{75}{8}$$

3. **Ans (B)**

$$A + C + B = 3B = 180^\circ$$

$$A + C = 120^\circ, B = 60^\circ$$

$$h = \ell(AB) - \ell(BC) \Rightarrow h = \frac{h}{\sin A} - \frac{h}{\sin C}$$

$$\Rightarrow \sin C - \sin A = \sin A \cdot \sin C$$

$$\Rightarrow 2 \sin\left(\frac{C-A}{2}\right) \cos\left(\frac{C+A}{2}\right)$$

$$= \frac{1}{2} \{ \cos(C-A) - \cos(C+A) \}$$

$$\Rightarrow \sin\left(\frac{C-A}{2}\right) = \frac{1}{2} \left\{ 1 - 2\sin^2\left(\frac{C-A}{2}\right) \right\} + \frac{1}{4}$$

$$\Rightarrow \sin^2\left(\frac{C-A}{2}\right) + \sin\left(\frac{C-A}{2}\right) - \frac{3}{4} = 0$$

$$\Rightarrow \sin\left(\frac{C-A}{2}\right) = \frac{1}{2}$$

4. **Ans (B)**

Let (h,k) be the locus

$$\text{So, } 3h = a \cos \theta + b \sin \theta \quad \dots(i)$$

$$\text{and } 3k = a \sin \theta - b \cos \theta + 1$$

$$\Rightarrow 3k - 1 = a \sin \theta - b \cos \theta \quad \dots(ii)$$

Now $(i)^2 + (ii)^2$ eliminates ' θ '

SECTION-I (ii)

5. **Ans (A,B,D)**

$$x^3 + ax^2 + bx + c = (x - \alpha)(x - \beta)(x - \gamma) \text{ where}$$

$$\alpha, \beta, \gamma \in \mathbb{Z}$$

$$-2009 = (1 - \alpha)(1 - \beta)(1 - \gamma)$$

$$-7 \times 7 \times 41 = (1 - \alpha)(1 - \beta)(1 - \gamma)$$

$$\text{Compare } \alpha = 8, \beta = 8, \gamma = 42$$

$$a = -58$$

6. **Ans (A,C)**

We have,

$$x^2 - (x+y) \cdot \left(\frac{y-mx}{c} \right) = 0 \quad (\text{using principle of homogenising})$$

Now, put (coefficient of x^2) + (coefficient of y^2) = 0

$$\Rightarrow 1 = c + m$$

$$\Rightarrow 1 = c + m$$

$$\text{Also, } (y-1) + m(1-x) = 0$$

So, fixed point (1, 1)

7. Ans (A,D)

A(2,0)

B(0,2)

$$2x + 3y + 1 = 0$$

|PA - PB| is minimum

$$\Rightarrow PA = PB$$

intersection point of $2x + 3y + 1 = 0$

$$\& y - 1 = x - 1$$

$$\Rightarrow \left(-\frac{1}{5}, -\frac{1}{5}\right)$$

|PA - PB| is maximum

$\Rightarrow$ intersection point of

$$2x + 3y + 1 = 0 \& x + y = 2$$

$$\Rightarrow (7, -5)$$

8. Ans (B,C,D)

Let a_n denotes the side of the square S_n then

$$a_n = \sqrt{2}a_{n+1}$$

$$\Rightarrow \frac{a_{n+1}}{a_n} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow a_n = a_1 \left(\frac{1}{\sqrt{2}}\right)^{n-1} = 10 \left(\frac{1}{\sqrt{2}}\right)^{n-1}$$

Now, we must have $a_n^2 < 1$

$$\Rightarrow 100 \left(\frac{1}{\sqrt{2}}\right)^{2n-2} < 1$$

$$\Rightarrow 2^n > 200 \Rightarrow n > 7$$

9. Ans (B,C)

Required line must pass through the mid-point of (2,0) and (0,4) and has slope 1 or -1

$$\text{So, } y - 2 = \pm(x - 1)$$

10. Ans (A,B,D)

The circumcircle of triangle passes through the mirror image of the orthocentre about the side. Also centroid divides orthocentre & circumcentre in 2 : 1 ratio.

11. Ans (A,C,D)

Let the locus of G be (h,k)

Also, A(a,0) and B(0,b)

$$\frac{a}{3} = h \text{ and } \frac{b}{3} = k$$

$$\Rightarrow (3h)^2 + (3k)^2 = 81$$

$$\Rightarrow x^2 + y^2 = 9$$

12. Ans (B,C,D)

We take the general element of second set, counting backward, as

$$t_n = 498 + (n - 1)(-5)$$

Also, for the first set, counting forward as

$$t'_m = 1 + (m - 1) \times 3$$

$$\text{Hence, } t_n = t'_m$$

$$\Rightarrow 505 - 5n = 3m$$

$$\Rightarrow 5(101 - n) = 3m$$

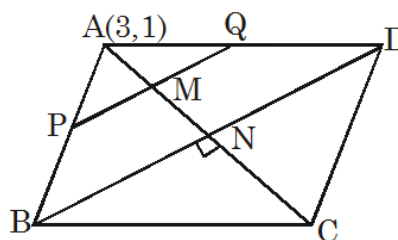
we want the least n such that (101 - n) is a multiple of 3.

$$\text{Hence, } n = 2$$

$$\text{so, } t_2 = 498 - 5 = 493$$

SECTION-II

1. Ans (1.00)



Let P & Q are mid points of side AB & AD respectively $PQ \parallel BD$, $AN \perp BD$

$$AN = \left| \frac{9 - 4 + 5}{5} \right| = 2$$

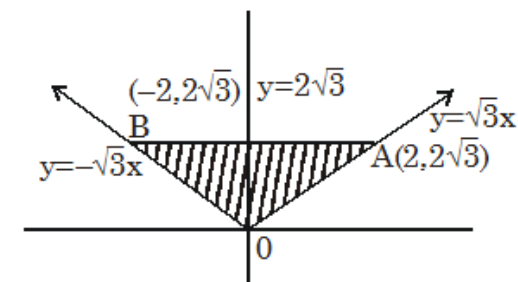
$$AM = 1$$

Equation of PQ $\Rightarrow 3x - 4y + \lambda = 0$ PQ is at distance 1 unit from BD $\Rightarrow \lambda = 0, 10$

$$\because \text{PQ is at distance 1 unit from A} \Rightarrow \lambda = 0$$

$$PQ \text{ is } 3x - 4y = 0$$

2. Ans (7.00)



$$\text{area} = \frac{1}{2} (4) (2\sqrt{3}) = 4\sqrt{3}$$

3. **Ans (6.00)**

Given $a(1 + r + r^2) = 21 \quad \dots(1)$

& $a^2(1 + r^2 + r^4) = 189 \quad \dots(2)$

$$\frac{\text{Eq. (2)}}{\text{Eq. (1)}^2} \Rightarrow \frac{r^2 - r + 1}{r^2 + r + 1} = \frac{3}{7}$$

$$\Rightarrow 7r^2 - 7r + 7 = 3r^2 + 3r + 3$$

$$\Rightarrow 4r^2 - 10r + 4 = 0$$

$$\Rightarrow 2r^2 - 5r + 2 = 0$$

$$\Rightarrow 2r^2 - 4r - r + 2 = 0$$

$$\Rightarrow 2r(r - 2) - 1(r - 2) = 0$$

$$\Rightarrow r = \frac{1}{2} \quad (\because r \notin \mathbb{I})$$

$$\therefore (1) \Rightarrow a \left(1 + \frac{1}{2} + \frac{1}{4} \right) = 21 \Rightarrow a = 12$$

$$\therefore S = \frac{a}{1 - r} = \frac{12}{1 - \frac{1}{2}} = 24 \Rightarrow \frac{S}{4} = 6$$

4. **Ans (4.00)**

$$\underbrace{(4\cos^2 x + \sec^2 x)}_{\geq 4} + \underbrace{(\tan^2 x + \cot^2 x)}_{\geq 2} = 6$$

$$\therefore \cos^2 x = \frac{1}{2} \text{ and } \tan^2 x = 1$$

$$\therefore \cos x = \pm \frac{1}{2}$$

Hence 4 solution

5. **Ans (7.00)**

$$\Delta = \begin{vmatrix} \gamma^2 & \beta^2 & 2\beta\gamma - \alpha^2 \\ 2\alpha\gamma - \beta^2 & \alpha^2 & \gamma^2 \\ \alpha^2 & 2\alpha\beta - \gamma^2 & \beta^2 \end{vmatrix}$$

$$= \begin{vmatrix} \gamma & \beta & \alpha \\ \alpha & \gamma & \beta \\ \beta & \alpha & \gamma \end{vmatrix} \begin{vmatrix} \gamma & \alpha & \beta \\ \alpha & \beta & \gamma \\ -\beta & -\gamma & -\alpha \end{vmatrix}$$

$$= (\alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma)^2$$

6. **Ans (4.23 or 4.24)**

Equation of line through (1,2) and parallel to y

= x is

$$y - 2 = x - 1 \Rightarrow y = x + 1$$

For intersection with $y = 2x + 3$;

$$x = -2 \text{ and } y = -1$$

Hence, required distance

$$= \sqrt{9+9} = 3\sqrt{2}$$

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,D	A,C,D	A,C,D	A,B	A,C,D	B,C,D	A,B	A,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	D	B	C	B				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	90.00	80.00	40.00	80.00	20.00	52.00		

PART-2 : CHEMISTRY

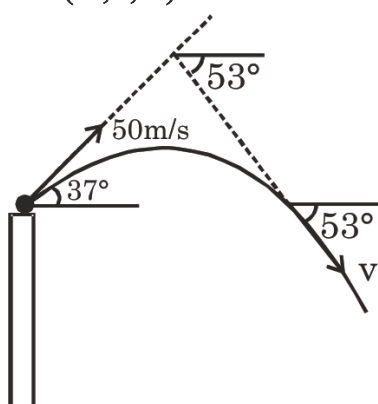
SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B	B,C	A,B,D	A,C,D	A,C	A,B	A,B,C,D	A,C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	D	C	C	D				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	4.00	6.00	3.00	5.00	15.00	7.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	B,D	B,C	A,D	A,C,D	A,D	A,B,C,D	B,C,D	A
SECTION-I (ii)	Q.	9	10	11	12				
	A.	B	D	C	B				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	8.00	2.00	142.00	5.00	10.00	4.75		

HINT – SHEET**PART-1 : PHYSICS****SECTION-I (i)**

1. Ans (A,B,D)



Equation of trajectory

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

$$y = x \tan 37^\circ - \frac{gx^2}{2(50)^2 \cos^2 37^\circ}$$

$$y = \frac{3x}{4} - \frac{x^2}{320}$$

For particle to move perpendicular

$$50 \cos 37^\circ = v \cos 53^\circ$$

$$\Rightarrow v = \frac{200}{3}$$

$$\Rightarrow v_y = -v \sin 53^\circ = -\frac{160}{3}$$

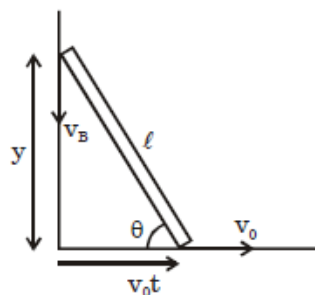
$$\Rightarrow v_y = u_y + a_y t$$

$$-\frac{160}{3} = 30 - 10t \Rightarrow t = \frac{25}{3} \text{ sec}$$

 $v_{\min} = 40 \text{ m/s}$ at top most point.

$$H_{\max} = \frac{u^2 \sin^2 \theta}{2g} = \frac{30 \times 30}{20} = 45 \text{ m}$$

2. Ans (A,C,D)



$$v_B \sin \theta = v_0 \cos \theta$$

$$v_B \sin \theta = v_0 \left(\frac{v_0 t}{l} \right)$$

$$v_B = v_0 \cot \theta$$

$$v_B = \frac{v_0 (v_0 t)}{y}$$

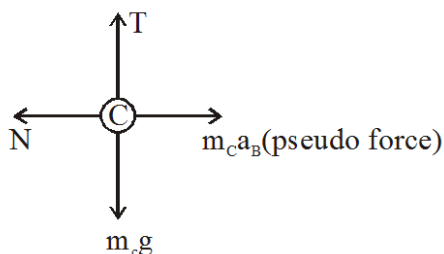
$$y^2 + (v_0 t)^2 = l^2$$

$$\frac{v^2}{l^2} + \frac{v_0^2 t^2}{l^2} = 1$$

$$\frac{y^2}{l^2} + \frac{t^2}{\left(\frac{l}{v_0} \right)^2} = 1$$

3. Ans (A,C,D)

In frame of B

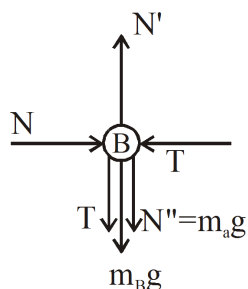


$$a_{cy} = \underbrace{a_A + a_B}_{a_{A/B}} \dots\dots(1)$$

$$\left. \begin{aligned} N &= 5a_B \dots\dots\dots(2) \\ 50 - T &= 5a_{cy} \dots\dots(3) \end{aligned} \right\} \text{FBD of C}$$

$$T + 300 a_B = 30 (a_A + a_B) \dots\dots(4) \text{ FBD of A}$$

In Ground Frame



$$T - N = 5a_B \dots\dots(5) \text{ FBD of B}$$

From Eq. (3), (4) and (1)

$$50 + 30a_B = 35a_{cy} \dots\dots(5)$$

from eqⁿ (1) and (4)

$$50 = 50 a_{cy} - 10a_B$$

from eq (5) and (6)

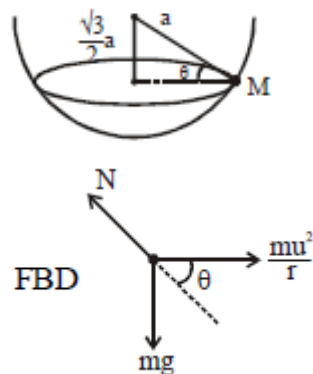
$$\left. \begin{aligned} a_{cy} &= 4\text{m/s}^2 \\ a_B &= 4\text{m/s}^2 \end{aligned} \right\} \text{ Putting this eq}^n(1) \quad a_A = 1 \text{ m/s}^2 \text{ (A)}$$

from eqⁿ (2) from eqⁿ (3)

$$N = 15 \text{ N (D)} \quad T = 30 \text{ N (C)}$$

$$N' = \left\{ m_B + m_A + \frac{T}{g} \right\} g = 380 \text{ N}$$

4. Ans (A,B)



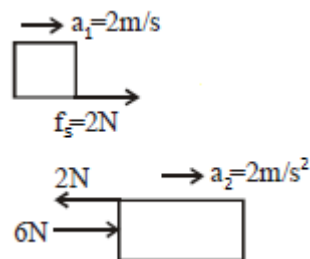
$$\theta = 60^\circ$$

$$(i) r = a \cos 60^\circ = \frac{a}{2}$$

$$(ii) \tan \theta = \frac{mg}{\frac{mu^2}{r}}$$

$$\sqrt{3} = \frac{ag}{2u^2} \Rightarrow u = \sqrt{\frac{ag}{2\sqrt{3}}}$$

5. Ans (A,C,D)

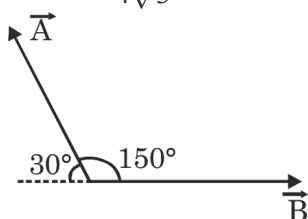
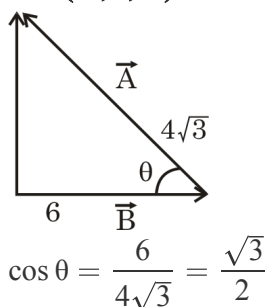


Work done by friction on A $\neq 0$

$$V = 2 \times 2 = 4 \text{ m/s}$$

$$\text{K.E. A} = \frac{1}{2} \times 1 \times 4^2 = 8 \text{ J}$$

6. Ans (B,C,D)



$$\theta = 30^\circ$$

7. Ans (A,B)

(C) angular momentum about centre of mass

not conserve because tension force provide

torque

$$(D) \frac{d\vec{L}}{dt} = \text{torque} = \vec{r} \times \vec{F}$$

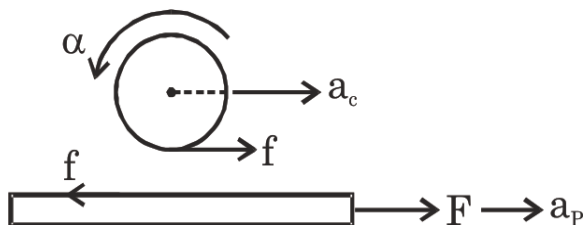
about centre of mass

$$\vec{r} \Rightarrow -\hat{i}$$

$$\vec{F} \Rightarrow -\hat{j} \quad (\text{tension force})$$

$$\vec{\tau} = -(\hat{i} \times \hat{j}) = -\hat{k}$$

8. Ans (A,C)

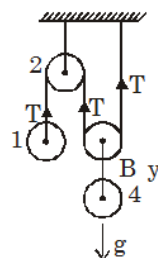


$$a_c + \alpha R = a_p$$

Friction force f is static

SECTION-I (ii)

9. Ans (D)



$$a_4 = g, a_2 = 0$$

$$-Ts_1 - 2Ts_3 = 0$$

$$a_1 + 2a_3 = 0$$

$$mg - G = ma_1$$

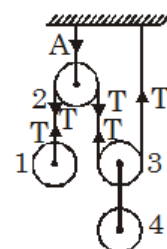
$$mg - 2T = ma_3$$

$$g - \frac{T}{m} + 2 \left(g - \frac{2T}{m} \right) = 0$$

$$3g = \frac{5T}{m} \Rightarrow T = \frac{3mg}{5}$$

$$a_1 = \frac{2g}{5} = 4m/s^2, a_3 = -\frac{g}{5} = -2m/s^2$$

10. Ans (B)



$$s_3 = s_4$$

$$-Ts_1 + 2Ts_2 - 2s_3 = 0$$

$$a_1 + 2a_3 = 2a_2$$

$$mg - T = ma_1$$

$$2T + mg = ma_2$$

$$2mg - 2T = 2ma_3$$

$$g - \frac{T}{m} + 2 \left(g - \frac{T}{m} \right) = 2 \left(\frac{2T}{m} + g \right)$$

$$g = \frac{7T}{m} \Rightarrow T = \frac{mg}{7} \Rightarrow a_1 = \frac{6g}{7}, a_2 = \frac{9g}{7},$$

$$a_3 = \frac{6g}{7}$$

11. Ans (C)

$$mg \sin \theta = 60 \text{ N}$$

$$f_{s \max} = \mu mg \cos \theta = 80 \text{ N}$$

12. Ans (B)

$$f_{s\max} = \mu mg \cos \theta = \frac{1}{2} \times 10 \times 10 \times \frac{4}{5} = 40 \text{ N}$$

$$mg \sin \theta = 60 \text{ N}$$

$$(P) t = 4 \text{ s}$$

$$F_1 + F_2 = 60 \text{ N} \Rightarrow f_s = 0$$

$$(Q) t = 7 \text{ s}$$

$$F_1 + F_2 = 90 \text{ N} \Rightarrow f_s = 30$$

$$(R) t = 10 \text{ s}$$

$$F_1 + F_2 = 120 \text{ N} \Rightarrow f = f_k = 40 \text{ N}$$

$$(S) t = 15 \text{ s}$$

$$F_1 + F_2 = 170 \text{ N} \Rightarrow f = f_k = 40 \text{ N}$$

SECTION-II

2. Ans (80.00)

For block A

$$40 \sin 37^\circ - T = 4a \dots (1)$$

For block B

$$T - 200 = 2a \dots (2)$$

(1) & (2)

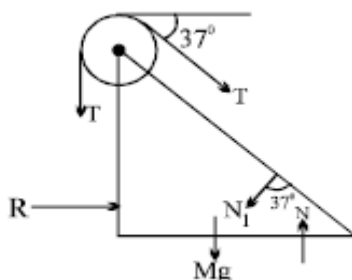
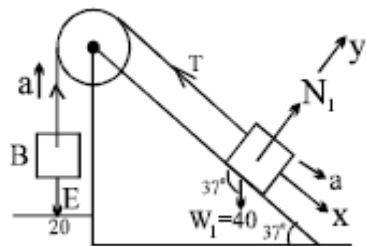
$$T = \frac{64}{3} \text{ N}$$

consider inclined as a system

$$R = N_1 \sin 37^\circ - T \cos 37^\circ$$

$$= (40 \cos 37^\circ) \sin 37^\circ - T \cos 37^\circ$$

$$= (40) \frac{4}{5} \times \frac{3}{5} - \frac{64}{3} \times \frac{4}{5} = \frac{32}{15} \text{ N}$$



4. Ans (80.00)

we have given

$$v = 3x^2 - 2x$$

we know that $a = v \cdot \frac{dv}{dx}$

$$\vec{a} = (3x^2 - 2x) \frac{d}{dx} (3x^2 - 2x)$$

$$a|_{x=2} = [3x^2 - 2x] \cdot [6x - 2]$$

$$= [3(2)^2 - 2(2)] \cdot [6 \times 2 - 2]$$

$$a|_{x=2} = [12 - 4] \cdot [12 - 2]$$

$$\vec{a} = 8 \times 10 = 80 \text{ m/s}^2$$

5. Ans (20.00)

$$W_{mg} + W_{\text{air friction}} = K_f - K_i$$

$$-mgh + W_{af} = K_f - K_i$$

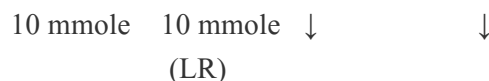
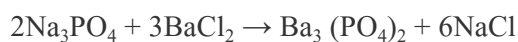
$$-1 \times 10 \times 18 + W_{af} = 0 - \frac{1}{2} \times 1 \times (20)^2$$

$$W_{af} = -20 \text{ J} \Rightarrow \text{Energy loss} = 20 \text{ J}$$

PART-2 : CHEMISTRY

SECTION-I (i)

4. Ans (A,C,D)



Left mmols

$$\left(10 - \frac{2}{3} \times 10\right) \quad 0 \quad \text{Out of solution}$$

$$\left(\frac{6}{3} \times 10\right) \text{ mmols}$$

5. Ans (A,C)

Let

$$\text{K}_2\text{C}_2\text{O}_4 = x \text{ mole}$$

$$\text{KHC}_2\text{O}_4 = y \text{ mole}$$

$$y \times 1 = 10 \times 0.05 \times 10^{-3}$$

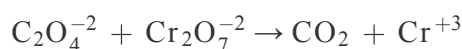
$$y = 5 \times 10^{-4}$$

$$\text{with KMnO}_4 \quad 2x + 2y = 0.6 \times 10^{-3} \times 5$$

$$x = 10 \times 10^{-4}$$

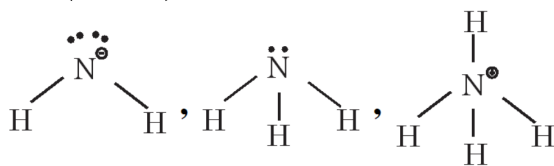
$$\boxed{\frac{x}{y} = \frac{2}{1}}$$

in 20 ml solution



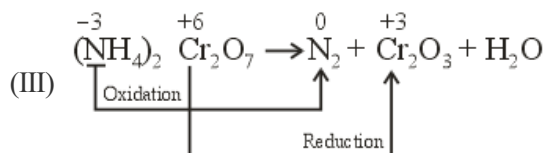
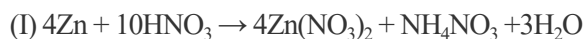
$$\text{Milliequivalent } 3 \times 2 = 6 \text{ of } \text{Cr}_2\text{O}_7^{2-}$$

8. Ans (A,C,D)



SECTION-I (ii)

9. Ans (D)



$$n_f = 6$$

SECTION-II

4. Ans (5.00)

$$[\text{K}^+] = \frac{M_1 V_1 \times x + M_2 V_2}{x \times V_1 + V_2}$$

$$0.06 = \frac{0.1 \times 5 \times x + 0.02 \times 25}{x \times 5 + 25}$$

$$x = 5$$

6. Ans (7.00)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B,D)

$$m + m^2 = 6 \quad m^3 = a$$

$$(m + m^2)^3 = 216$$

$$m^3 + m^6 + 3m^3(m + m^2) = 216$$

$$a + a^2 + 18a = 216$$

$$a^2 + 19a - 216 = 0$$

$$(a + 27)(a - 8) = 0$$

$$a = 8, -27$$

2. Ans (B,C)

For concurrency

$$\begin{vmatrix} 1 & 1 & -1 \\ m-1 & m^2-7 & -5 \\ m-2 & 2m-5 & 0 \end{vmatrix} = 0$$

$$\Rightarrow 5(2m-5) - 5(m-2) - [(m-1)(2m-5) - (m-2)(m^2-7)] = 0$$

$$\Rightarrow 10m - 25 - 5m + 10 - 2m^2 + 7m - 5 + m^3 - 2m^2 - 7m + 14 = 0$$

$$\Rightarrow m^3 - 4m^2 + 5m - 6 = 0$$

$$\Rightarrow (m-3)(m^2 - m + 2) = 0$$

$m = 3$ is the only real root

But for $m = 3$ the given line becomes parallel. Thus the given lines can only be parallel for $m = 3$.

3. Ans (A,D)

For unique solution of system of given equation

$$\begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix} \neq 0$$

& for new system of homogeneous equation

$$D = \begin{vmatrix} q+r & r+p & p+r \\ r+p & p+q & r+p \\ p+q & q+r & r+p \end{vmatrix} \neq 0$$

$$\Rightarrow x = y = z = 0$$

for new system of equation

$\therefore$ it is consistent & gives unique solutions.

4. Ans (A,C,D)

$$a \log_c 2, b^3, \frac{c}{b^3}, \frac{64}{\log_c 2^a} \rightarrow \text{H.P}$$

H.M. < G.M.

$$b^3 < \sqrt{a \cdot \log_c 2 \cdot \frac{c}{b^3}} \quad \dots(i)$$

$$\frac{c}{b^3} < \sqrt{b^3 \cdot \frac{64}{a \cdot \log_c 2}} \quad \dots(ii)$$

$$\Rightarrow c < \sqrt{64c}$$

$$\Rightarrow c(c - 64) < 0 \Rightarrow c \in (0, 64)$$

Due to domain $\Rightarrow c > 0, c \neq 1$

5. **Ans (A,D)**

$$\tan^3 x - 1 - 5 \tan x (\tan x - 1) = 0$$

$$\Rightarrow (\tan x - 1)(\tan^2 x - 4 \tan x + 1) = 0$$

$$\Rightarrow \tan x = 1$$

$$\& \tan^2 x - 4 \tan x + 1 = 0 \Rightarrow \tan x = 2 \pm \sqrt{3}$$

check options

6. **Ans (A,B,C,D)**

$$\sqrt{2} - 1 \leq \tan x \leq \sqrt{2} + 1$$

$$\Rightarrow n\pi + \frac{\pi}{8} \leq x \leq \frac{3\pi}{8} + n\pi$$

Put $n = 0, 1, -1, -2$

7. **Ans (B,C,D)**

$$\sin A = 7k, \sin B = 25k, \sin C = 24k$$

$$\Rightarrow \sin^2 B = \sin^2 A + \sin^2 C$$

$$\Rightarrow b^2 = a^2 + c^2 \Rightarrow b^2 - c^2 = a^2$$

$\Rightarrow B$ is correct.

$$(C) \quad a^{\log a - \log b} \times b^{\log a - \log b} \times b^{\log b - \log c} \times c^{\log b - \log c} \\ \times a^{\log c - \log a} \times a^{\log c - \log a}$$

$$\Rightarrow a^{\log c - \log b} \times b^{\log a - \log c} \times c^{\log b - \log a}$$

$$= \frac{a^{\log c} \times b^{\log a} \times c^{\log b}}{a^{\log b} \times b^{\log c} \times c^{\log a}} = 1$$

8. **Ans (A)**

$$\text{slope of req. line} = \frac{-1}{\left(\frac{1}{-1}\right)} = 1$$

$$\therefore \text{line is } y - 1 = x - 3$$

$$\Rightarrow y = x - 2$$

SECTION-I (ii)

9. **Ans (B)**

$$2a^2 + 4b^2 + c^2 - 4ab - 2ac = 0$$

$$\Rightarrow (a - 2b)^2 + (a - c)^2 = 0$$

$$\Rightarrow a = 2b, a = c \Rightarrow a = 2b = c$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{1}{4}$$

$$\text{similarly } \cos B = \frac{7}{8}, \cos C = \frac{1}{4}$$

also as $a = c \Rightarrow A = C$

$$\Rightarrow \sin(A - C) = 0$$

10. **Ans (D)**

$$2a^2 + 4b^2 + c^2 - 4ab - 2ac = 0$$

$$\Rightarrow (a - 2b)^2 + (a - c)^2 = 0$$

$$\Rightarrow a = 2b, a = c \Rightarrow a = 2b = c$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{1}{4}$$

$$\text{similarly } \cos B = \frac{7}{8}, \cos C = \frac{1}{4}$$

also as $a = c \Rightarrow A = C$

$$\Rightarrow \sin(A - C) = 0$$

11. **Ans (C)**

$$(I) \quad e^x + e^{2x} - 5 = 0$$

$$\Rightarrow e^x = \frac{-1 + \sqrt{21}}{2} \text{ or } \frac{-1 - \sqrt{21}}{2} \text{ (reject)}$$

$$(II) \quad \frac{a}{1-r} = 4 \quad \dots\dots(1)$$

$$\frac{a^2}{1-r^2} = 8 \quad \dots\dots(2)$$

Put a in equation (2)

$$\frac{16(1-r)(1-r)}{(1-r)(1+r)} = 8 \Rightarrow 2 - 2r = 1 + r$$

$$\Rightarrow 2 - 2r = 1 + r$$

$$\Rightarrow r = \frac{1}{3}$$

$$(III) \quad x^4 + \sqrt{x^4 + 10} = 10$$

$$x^4 + 10 + \sqrt{x^4 + 10} = 20$$

$$\text{Let } \sqrt{x^4 + 10} = t, t^2 + t = 20$$

$$t = -5 \text{ or } t = 4$$

reject

$$\therefore x^4 + 10 = 16 \Rightarrow x^4 = 6$$

$\Rightarrow$ two real roots.

$$(IV) \quad \log_2(1 + \sqrt{7x - x^2 - 6}) > 0$$

$$\Rightarrow 7x - x^2 - 6 > 0$$

$$x^2 - 7x + 6 < 0$$

$$x \in (1, 6)$$

12. Ans (B)

(I) $e^x + e^{2x} - 5 = 0$

$$\Rightarrow e^x = \frac{-1 + \sqrt{21}}{2} \text{ or } \frac{-1 - \sqrt{21}}{2}$$

(reject)

(II) $\frac{a}{1-r} = 4 \dots\dots(1)$

$$\frac{a^2}{1-r^2} = 8 \dots\dots(2)$$

Put a in equation (2)

$$\frac{16(1-r)(1-r)}{(1-r)(1+r)} = 8 \Rightarrow 2 - 2r = 1 + r$$

$$\Rightarrow 2 - 2r = 1 + r$$

$$\Rightarrow r = \frac{1}{3}$$

(III) $x^4 + \sqrt{x^4 + 10} = 10$

$$x^4 + 10 + \sqrt{x^4 + 10} = 20$$

$$\text{Let } \sqrt{x^4 + 10} = t, t^2 + t = 20$$

$$t = -5 \text{ or } t = 4$$

reject

$$\therefore x^4 + 10 = 16 \Rightarrow x^4 = 6$$

$\Rightarrow$ two real roots.

(IV) $\log_2(1 + \sqrt{7x - x^2 - 6}) > 0$

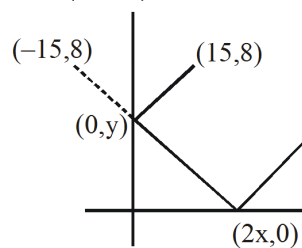
$$\Rightarrow 7x - x^2 - 6 > 0$$

$$x^2 - 7x + 6 < 0$$

$$x \in (1, 6)$$

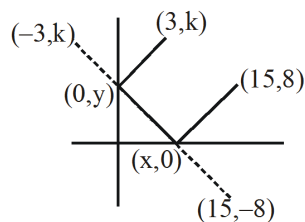
SECTION-II

1. Ans (8.00)



$$-\frac{15}{2x} + \frac{8}{y} = 1$$

$$-15y + 16x = 3xy \dots(1)$$

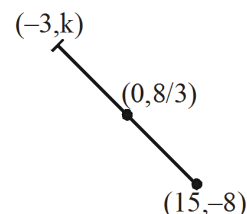


$$\frac{15}{x} - \frac{8}{y} = 1$$

$$15y - 8x = xy \dots(2)$$

$$(1) + (2) \quad 8x = 3xy$$

$$\therefore y = \frac{8}{3}$$



$$\frac{k - \frac{8}{3}}{-3} = \frac{\frac{8}{3} + 8}{0 - 15}$$

$$\Rightarrow \frac{3k - 8}{1} = \frac{32}{5}$$

$$\Rightarrow k = \frac{24}{5}$$

2. Ans (2.00)

$\sqrt{3}$, H, $\sqrt{2}$ in H.P.

$$\Rightarrow H = \frac{2\sqrt{6}}{\sqrt{2} + \sqrt{3}}$$

$$\Rightarrow \frac{H}{\sqrt{2}} = \frac{2\sqrt{3}}{\sqrt{2} + \sqrt{3}}, \quad \frac{H}{\sqrt{3}} = \frac{2\sqrt{2}}{\sqrt{2} + \sqrt{3}}$$

Using componendo dividendo

$$\frac{H + \sqrt{2}}{H - \sqrt{2}} + \frac{H + \sqrt{3}}{H - \sqrt{3}} = 2$$

3. Ans (142.00)

$$x^3 + ax + 2 = 0 \quad \dots(1)$$

$$\& x^4 + ax^2 + 6 = 0 \quad \dots(2)$$

multiply equation (1) by x & subtract from (2)

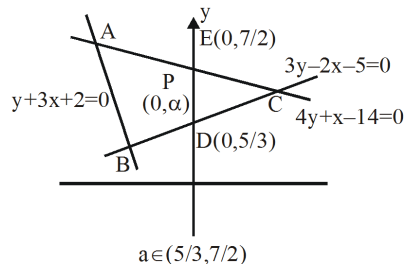
$$\Rightarrow x = 3 \Rightarrow \alpha = 3$$

therefore common root is 3, put $x = 3$ in equation (1)

$$\therefore (1) \Rightarrow 27 + 3a + 2 = 0 \Rightarrow a = -\frac{29}{3}.$$

$$\therefore |\alpha + 15a| = |3 - 145| = 142.$$

4. Ans (5.00)



5. Ans (10.00)

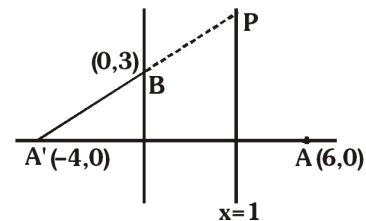
$$\begin{vmatrix} \log_3 2^9 & \log_{2^2} 3 \\ \log_3 2^3 & \log_{2^2} 3^2 \end{vmatrix} \begin{vmatrix} \log_2 3 & \log_{2^3} 3 \\ \log_3 2^2 & \log_3 2^2 \end{vmatrix}$$

$$\text{Let } \log_2 3 = \begin{vmatrix} \frac{9}{x} & \frac{x}{2} \\ \frac{3}{x} & x \end{vmatrix} \begin{vmatrix} x & \frac{x}{3} \\ \frac{2}{x} & \frac{2}{x} \end{vmatrix}$$

$$= \left(9 - \frac{3}{2}\right) \left(2 - \frac{2}{3}\right) = 10$$

6. Ans (4.75)

Let A' be the image of A in $x = 1$



$$PA - PB = PA' - PB \leq A'B$$

$$\therefore (PA' - PB)_{\max} = A'B$$

$$\therefore \text{Equation of } A'B \text{ is } y = \frac{3}{4}(x + 4)$$

$$4y = 3x + 12 \quad \therefore P \text{ is } \left(1, \frac{15}{4}\right)$$